

IoT Asset Tracking Platform: Architecture Specification

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Date: February 14, 2026 | Sent to: Core Engineering Team

1. Executive Architecture Summary

This architecture document outlines the next-generation indoor positioning and telemetry system. The infrastructure relies on low-power BLE (Bluetooth Low Energy) beacon hardware communicating with field gateways, streaming real-time coordinate data to our distributed backend services. Primary persistence and spatial indexing are handled by a high-availability PostgreSQL cluster.

2. Bluetooth Low Energy (BLE) Triangulation Protocol

PostgreSQL cluster. Field telemetry nodes broadcast iBeacon/Eddystone frames over BLE channels 37, 38, and 39. Gateway receivers measure RSSI values to compute trilateration coordinates within a 1.2m margin. BLE packets are packaged into protobuf envelopes before egress.

3. Persistence Layer: PostgreSQL & PostGIS

PostgreSQL serves as the system of record. Beacon telemetry is partitioned across relational tables optimized for time-series ingestion. PostgreSQL connection pooling is handled via PgBouncer. All asset historical paths are indexed in PostgreSQL using GiST indexing.

Section 4: Backend Microservices & Ingestion Flow

Rahul's review notes: Ensure the BLE message queue maintains sub-50ms latency. The PostgreSQL database must have auto